

Math 103

Class 10

1. **Probability** is a numerical measure between 0 and 1 that describes the likelihood that an event will occur. Probabilities closer to 1 indicate that the event is more likely to occur. Probabilities closer to 0 indicate that the event is less likely to occur.
2. $0 \leq P(E) \leq 1$ (boundaries for probabilities)
3. **Subjective Probability** uses values based on educated guesses, estimates, opinions and inexact information.
4. **Empirical Probability** relies on actual experience to determine the likelihood of outcomes.

$$P(E) = \frac{\text{frequency for the class}}{\text{total frequencies in the distribution}} = \frac{f}{n}$$

5. **Classical Probability** assumes that all outcomes in the sample space are equally likely to occur.

$$P(E) = \frac{n(E)}{n(S)} \quad \text{or} \quad \frac{FAV}{TOT} \quad \text{Express answer as a reduced fraction or decimal or percent.}$$

6. A **probability experiment** is a process that leads to well-defined results called outcomes.
7. An **outcome** is the result of a single trial of a probability experiment.
8. An **event** consists of one or more outcomes of a probability experiment.
9. A **sample space** is the set of all possible outcomes of a probability experiment.
10. Empirical Probability Example:

Out of 100 U.F. students randomly selected, 80 were taking Math 123, 10 were taking Math 140, 5 were taking Math 141, 4 were taking Math 132 and 1 was taking Math 320. Set up a frequency distribution and find the following probabilities.

- a.) a student is taking a 300 level class
- b.) a student is taking elementary statistics
- c.) a student is taking Math 141 or Math 132.
- d.) a student is taking Math 123 or Math 140

<u>Course</u>	<u>Frequency</u>	Answers:
Math 123	80	a.) 1/100
Math 140	10	b.) 4/5
Math 141	5	c.) 9/100
Math 132	4	d.) 9/10
Math 320	<u>1</u>	
	100	

11. If one card is drawn from a deck, find the probability of getting these results:

- a.) an ace ($4/52 = 1/13$)
- b.) a diamond ($13/52 = 1/4$)
- c.) a red card ($26/52 = 1/2$)
- d.) a red queen ($2/52 = 1/26$)

12. If a regular die is rolled one time, find these probabilities:

- a.) rolling a number greater than 2 ($4/6 = 2/3$)
- b.) rolling an even number ($3/6 = 1/2$)
- c.) rolling a prime number ($3/6 = 1/2$)

13. In an office there are 5 women and 4 men. If one person is selected, find the probability that the person is a woman.

(5/9)

14. 39 of 50 states are currently under court order to alleviate overcrowding and poor conditions in one or all of their prisons. If a state is selected at random, find the probability that it is currently under such a court order.

($39/50 = .78 = 78\%$)

15. The probability of rain is 20% for tomorrow. What does this mean?

(Given similar conditions, it has rained on 20 out of 100 days in the past.)

16. The math professor decided to make his last 3 questions on the probability test "true-false" questions. To guarantee that the answers do not follow any pattern; he lists all possible true-false combinations for three questions on slips of paper and then picks one at random from a hat. Make a list of all possible outcomes for this experiment.

TTT	FTT	TFT	FFT
TTF	FTF	TFF	FFF

- a.) What is the probability that all three will be false?
(1/8)
- b.) What is the probability that exactly 2 will be true?
(3/8)

17. **Complementary Events**

The complement of an event E is the set of outcomes in the sample space that are not included in the outcomes of event E. The complement of E is denoted by $\bar{E}$. (E bar)

The complement of rolling a 5 on a die would be rolling a 1, 2, 3, 4 or 6.

$P(5)$ is $1/6$

$P(1, 2, 3, 4 \text{ or } 6)$ is $5/6$. Note: $1/6 + 5/6 = 1$

The complement of rolling a sum of 7 with 2 dice is not rolling a sum of 7 with 2 dice.

$P(7) = 1/6$

$P(\text{not } 7) = 5/6$

The complement of drawing a club from a standard deck of cards is drawing a heart, a spade or a diamond.

$P(\text{club}) = 1/4$

$P(\text{not a club}) = 3/4$

The probability of an event and its complement will always add up to 1.